

How I would Blur Faces

(If I were to blur faces)

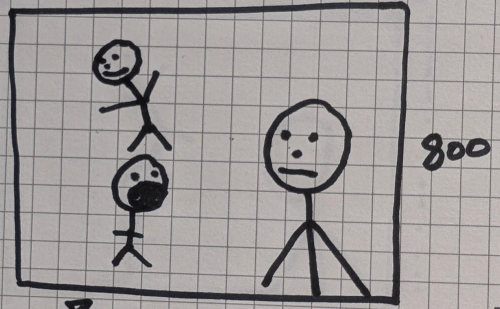
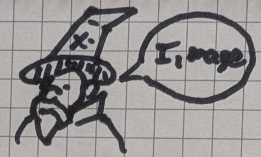
- Chris Rose

Choose a good model

- CNN
- Accurate (WIDER)
- Fast
- CPU-optimized

Ultimately you want to
train your own model

Image

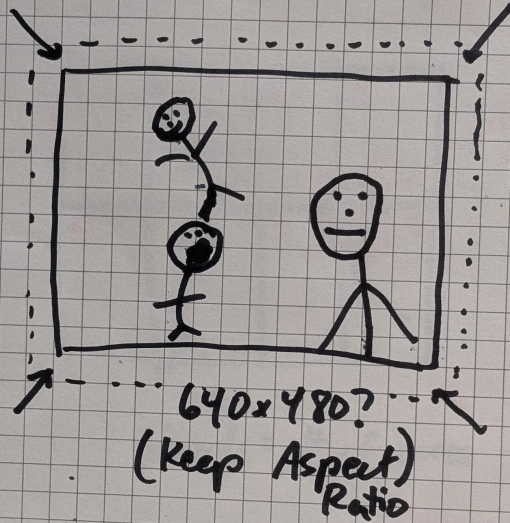


800

1200

Too Large

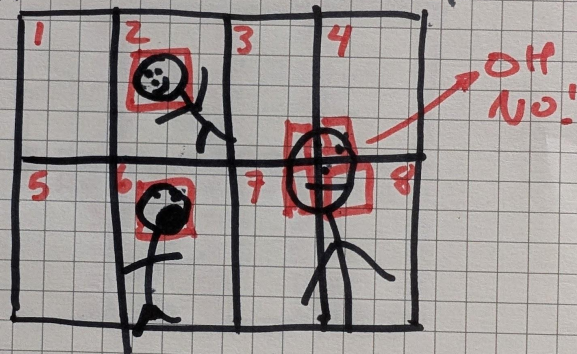
DownSCALE!



Lanczos

The smaller you scale
down, the faster

Sub-division! 
nice irregular rectangles



Threads = # of cores - 1

Especially
for spinlock'd
threads

Padding!

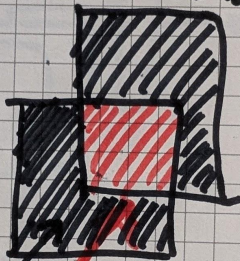


15% ?

Ideally $\frac{1}{2}$ of
average face
size.

Non-Maximum Suppression (NMS)

- Sort Boxes by Confidence
- Remove lower confidence boxes w/ significant overlap



boxes w/
significant
overlap

IoU

Intersection

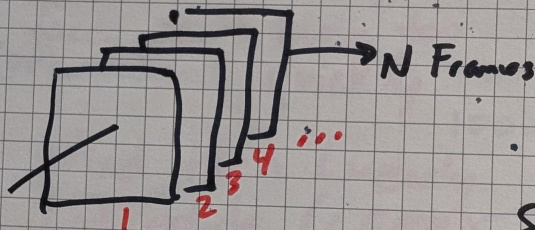
Union

= Ratio

80% ?
70%

Video

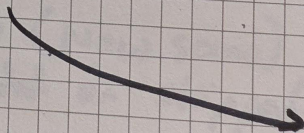
"Images over time"



GOAL:

AVOID DOING DETECTION
AS MUCH AS YOU CAN

Slow



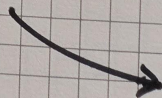
VIDEO TRICK #1

SKIP ~~X~~ Frames where
 Δt is low enough a
face can't move too
(or camera) quickly.

And then Linearly
Interpolate (LERP) $\rightarrow$ POSITION
(or exponential smooth) SIZE!
between.

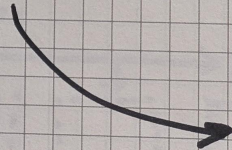
PROBLEMS:

- CUT IN VIDEO (New scene)
- New face added on skipped
Frame



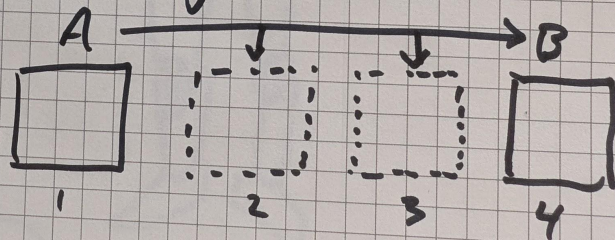
VIDEO TRICK #2

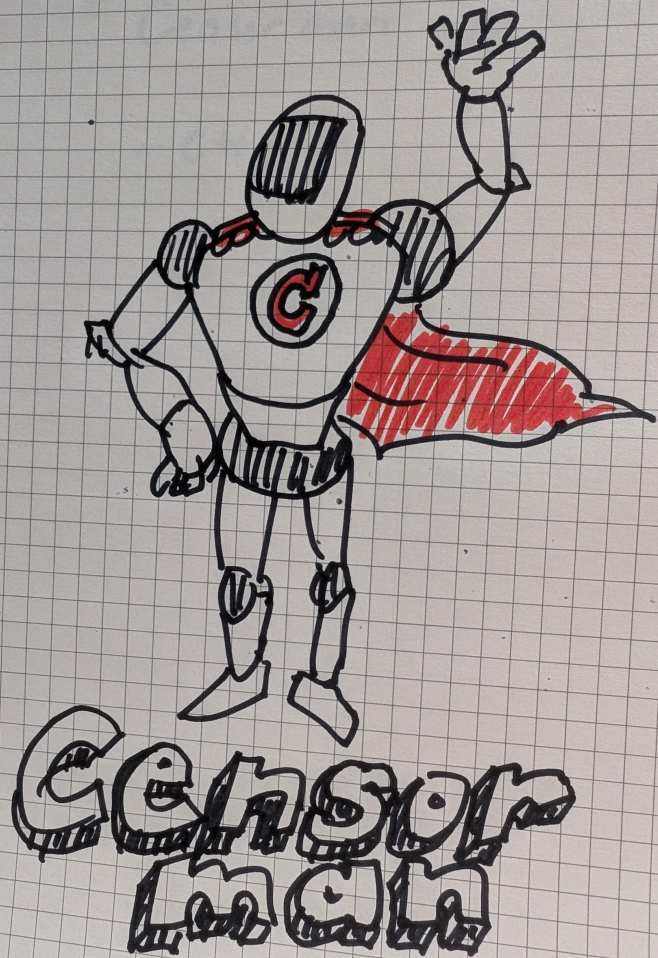
Do a first pass through
FRAME data and calculate
average pixel color in a
grid OR "heat map" of
pixel velocity over time
to pick frames smartly
and schedule for detection.



LERPING BOXES

- ① MATCH BOXES BY CONFIDENCE AND POSITION
- ② STORE VELOCITY AND ACCEL 2D VECTORS FOR EACH BOX FOR USE IN MATCHING OR PREDICTION
- ③ IF A BOX "disappears" for a frame, it may be best to LERP it because it is a model glitch.





Censor man

- C++ (≈ 2000 LOC)

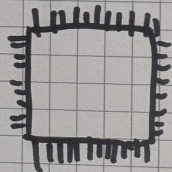
- .bin 6.2 MB

- Dependencies

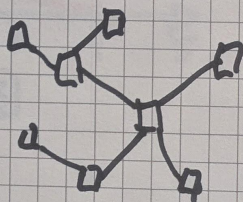
- STB-IMAGE.H
- STB-IMAGE-WRITE.H
- FFmpeg (x264)
- LIBFACEDETECTION

UNSOLICITED ADVICE:

- ① RESPECT THE
COMPUTER
- ② MINIMIZE
DEPENDENCIES

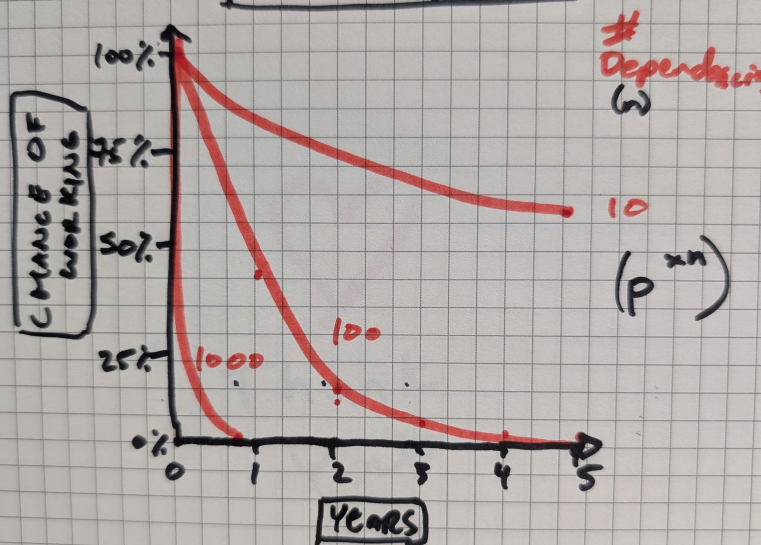


BILLIONS OF
OPERATIONS
PER SECOND

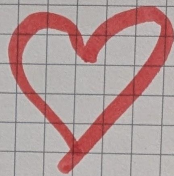


INTERNET
SPEED CLOSE
TO SPEED
OF LIGHT

MY PROGRAM'S CHANCE OF WORKING OVER TIME



* Assumed each dependency has a 99% chance of working after each year



Thank
you